



Yuze Zheng

Ph.D. Candidate in Chemistry, The
University of Chicago
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🌐 LinkedIn Profile

EDUCATION

•The University of Chicago

2024–present

Ph.D. in Chemistry, GPA: 3.88/4.00

Chicago, IL, USA

•Peking University

2020–2024

B.S. in Chemistry, GPA: 3.78/4.00, Top 10%

Beijing, China

RESEARCH EXPERIENCE

•Bozhi Tian Lab, Department of Chemistry, The University of Chicago

2024–present

Ph.D. Researcher

Chicago, IL, USA

- Developed light-responsive bioelectronic platforms at the interface of physical chemistry, materials chemistry, and bioelectronics.
- Investigated nanoplasmonic Au–TiO₂ multilayers to understand how dielectric environment, interfacial coupling, and nanoscale morphology govern photoinduced charge generation and transfer.
- Designed silicon-based MOS photocapacitor systems for wireless biomodulation through materials and interface engineering, including optimization of semiconductor–oxide–metal architectures and stimulation-interface coatings.
- Co-first author on manuscripts currently in revision at *Nature Photonics* and under review at *Science*; also published three first-author review and perspective articles and contributed a book chapter.

•Department of Chemistry, The University of Chicago

2024–2025

Teaching Assistant

Chicago, IL, USA

- Served as a Teaching Assistant for General Chemistry, leading discussion sections, grading laboratory reports and examinations, and supporting student learning through regular office hours.
- Designed and graded quizzes, guided laboratory activities, and maintained student performance records.

•Bingjun Xu Lab, College of Chemistry and Molecular Engineering, Peking University

2023–2024

Undergraduate Research Assistant

Beijing, China

- Developed a spatially resolved approach using bismuth reporter layers and operando Raman spectroscopy to map the active region within catalyst structures.
- Co-first author on a *JACS* publication identifying the catalytically active region in gas diffusion electrodes for electrochemical CO₂ reduction.

•Shu Hu Lab, Department of Chemical & Environmental Engineering, Yale University

2023

Research Assistant

New Haven, CT, USA

- Co-authored a *Nature Communications* paper demonstrating solar-driven selective electrochemical conversion of low-concentration dissolved carbon to fuels through engineered molecular flux interfaces.
- Engineered a high-resolution CO₂RR analysis system by integrating pH-sensitive optical mapping, flow-controlled in situ Raman spectroscopy, and COMSOL-based digital twins to uncover transport mechanisms and interfacial reactivity.

•Bingjun Xu Lab, College of Chemistry and Molecular Engineering, Peking University

2022–2023

Undergraduate Research Assistant

Beijing, China

–Conducted propane dehydrogenation experiments in a fixed-bed reactor and used GC, in situ FTIR, and O₂ pulse titration to quantify Ga species and investigate how Al site proximity influences catalyst structure and performance.

PUBLICATIONS

Research Articles

- Q. Sun^{*}, Y. Zheng^{*}, ..., X. Chang, B. Xu. *Identification of Active Region in Catalyst Layer on Gas Diffusion Electrode in Electroreduction of CO₂*. **JACS**, 2025.
- B. Liu, ..., Y. Zheng, ..., S. Hu. *Solar-driven selective conversion of millimolar dissolved carbon to fuels with molecular flux generation*. **Nature Communications**, 2025.

Review and Perspective Articles

- Y. Zheng, G. Yang, P. Li, B. Tian. *Bioelectric and physicochemical foundations of bioelectronics in tissue regeneration*. **Biomaterials**, 2025.
- G. Yang^{*}, Y. Zheng^{*}, B. Tian. *Perspective on Nano-Enabled Photostimulation Biointerfaces*. **ACS Nano**, 2025.
- Y. Zheng, G. Yang, B. Tian. *Soft electroceutical immunomodulation*. **National Science Review**, 2026.

Manuscripts in Revision / Under Review

- P. Li^{*}, ..., Y. Zheng^{*}, ..., B. Tian. *Self-Organized Nanoplasmonic Artificial Leaf for Hot-Carrier Bioelectronic Interfaces*. **Nature Photonics**, in revision.
- P. Li^{*}, G. Yang^{*}, Y. Zheng^{*}, ..., B. Tian. *Metal-Oxide-Semiconductor Augmented Skin for Safe and Universal Optoelectronic Therapies*. **Science**, under review.

Book Chapter

- G. Yang^{*}, Y. Zheng^{*}, P. Li, B. Tian. *Fabrication of Heterogeneous Silicon-Based Mesosstructures*. **Handbook of Porous Silicon**, 3rd ed. Accepted in principle.

^{*}Co-first authors.

TECHNICAL SKILLS

- **Programming and Data Analysis:** Python, MATLAB, Origin
- **Scientific Software and Design:** COMSOL Multiphysics, AutoCAD, ChemDraw, Adobe Illustrator
- **Microfabrication and Device Fabrication:** Photolithography, atomic layer deposition (ALD), physical vapor deposition (PVD), chemical vapor deposition (CVD), reactive ion etching (RIE), wet etching, sputtering, electroplating, laser cutting
- **Materials Characterization:** FTIR, Raman spectroscopy, fluorescence microscopy, HPLC, GC, SEM/EDS
- **Biological and Bioelectronic Models:** In vivo and ex vivo rat heart models, fibroblast and cardiomyocyte isolation, sciatic and hypoglossal nerve exposure in rats, rat dorsal root ganglion neuron culture
- **Languages:** Mandarin Chinese (native), English (full professional proficiency)

AWARDS AND HONORS

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| •Model Student of Academic Records
<i>Peking University</i> | Oct 2023 |
| •Third-Class Scholarship
<i>Peking University</i> | Oct 2023 |
| •Third-Class Scholarship
<i>Peking University</i> | Oct 2022 |
| •First Prize, National College Students Chemical Experiment Design Competition
<i>Chinese Chemical Society</i> | Aug 2022 |
| •Second-Class Scholarship
<i>Peking University</i> | Oct 2021 |
| •Qin Jin Scholarship
<i>Peking University</i> | Jul 2021 |